

SOLIDWORKS API (NET VBA C#) — Reference (206/337): IModeler (part 4 of 4)

SOLIDWORKS API Documentation (converted from CHM)

IModeler (part 4 of 4) (Type)

SOLIDWORKS API Help

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MergeCurves Method (IModeler)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : MergeCurves Method (IModeler)

Merges multiple curves into one curve.

.NET Syntax

Visual Basic (Declaration)

```
Function MergeCurves( _  
    ByVal CurveVar As System.Object _  
) As Curve
```

Visual Basic (Usage)



```
Dim instance As IModeler  
Dim CurveVar As System.Object  
Dim value As Curve  
  
value = instance.MergeCurves(CurveVar)
```

C#

```
Curve MergeCurves(  
    System.object CurveVar  
)
```

C++/CLI

```
Curve^ MergeCurves(  
& System.Object^ CurveVar  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

CurveVar
Array of [curves](#) to merge

Return Value

Newly created merged [curve](#) or Nothing if merge fails

Visual Basic for Applications (VBA) Syntax

See [Modeler::MergeCurves](#).

See Also

[IModeler Interface](#)
[IModeler Members](#)
[IModeler::IMergeCurves Method](#)

Availability

SOLIDWORKS 2006 FCS, Revision Number 14.0

SOLIDWORKS API Help

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ProjectCurveOnSurface Method (IModeler)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : ProjectCurveOnSurface Method (IModeler)

Projects a curve on a surface.

.NET Syntax

Visual Basic (Declaration)

```
Function ProjectCurveOnSurface( _  
    ByVal CurveIn As Curve, _  
    ByVal SurfaceIn As Surface, _  
    ByVal ProjDir As MathVector _  
) As Curve
```

Visual Basic (Usage)



```
Dim instance As IModeler  
Dim CurveIn As Curve  
Dim SurfaceIn As Surface  
Dim ProjDir As MathVector  
Dim value As Curve  
  
value = instance.ProjectCurveOnSurface(CurveIn, SurfaceIn, ProjDir)
```

C#

```
Curve ProjectCurveOnSurface(  
    Curve CurveIn,  
    Surface SurfaceIn,  
    MathVector ProjDir  
)
```

C++/CLI

```
Curve^ ProjectCurveOnSurface(  
& Curve^ CurveIn,  
& Surface^ SurfaceIn,  
& MathVector^ ProjDir  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

CurveIn
[Curve](#) to project
SurfaceIn
[Surface](#) on which to project curve
ProjDir
[Direction](#) which to project curve

Return Value

Projected [curve](#)

Visual Basic for Applications (VBA) Syntax

See [Modeler::ProjectCurveOnSurface](#).

Remarks

This method does not support non-spline curves. You must convert any non-spline input curves to splines before using this method.

See Also

[IModeler Interface](#)
[IModeler Members](#)
[ISurface::MakeIsoCurve Method](#)
[ISurface::IMakeIsoCurves Method](#)
[ISurface::MakeIsoCurve Method](#)

Availability

SOLIDWORKS 2006 FCS, Revision Number 14.0

ReplaceSurfaces Method (IModeler)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : ReplaceSurfaces Method (IModeler)

Replaces the surfaces of the given faces.

.NET Syntax

Visual Basic (Declaration)

```
Function ReplaceSurfaces( _  
    ByVal NFaces As System.Integer, _  
    ByVal FaceArray As System.Object, _  
    ByVal NewSurfArray As System.Object, _  
    ByVal SenseArray As System.Object, _  
    ByVal Tolerance As System.Double _  
    ) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModeler  
Dim NFaces As System.Integer  
Dim FaceArray As System.Object  
Dim NewSurfArray As System.Object  
Dim SenseArray As System.Object  
Dim Tolerance As System.Double  
Dim value As System.Boolean  
  
value = instance.ReplaceSurfaces(NFaces, FaceArray, NewSurfArray, SenseArray, Tolerance)
```

C#

```
System.bool ReplaceSurfaces(  
    System.int NFaces,  
    System.object FaceArray,  
    System.object NewSurfArray,  
    System.object SenseArray,  
    System.double Tolerance  
)
```

C++/CLI

```
System.bool ReplaceSurfaces(  
& System.int NFaces,  
& System.Object^ FaceArray,  
& System.Object^ NewSurfArray,  
& System.Object^ SenseArray,  
& System.double Tolerance  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

NFaces

Number of faces to have surfaces replaced

FaceArray

Array of [faces](#) to have the surfaces replaced

NewSurfArray

Array of [surfaces](#) to be replaced in these faces

SenseArray

Array of the senses of each surface in NewSurfArray

Tolerance

Tolerance

Return Value

True if operation is successful, false if not

Visual Basic for Applications (VBA) Syntax

See [Modeler::ReplaceSurfaces](#).

See Also

[IModeler Interface](#)
[IModeler Members](#)
[IModeler::IReplaceSurfaces2 Method](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 0.0

SOLIDWORKS API Help

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Restore Method (IModeler)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : Restore Method (IModeler)

Restores a temporary body object from the specified stream.

.NET Syntax

Visual Basic (Declaration)

```
Function Restore( _  
    ByVal StreamIn As System.Object _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IModeler  
Dim StreamIn As System.Object  
Dim value As System.Object
```

```
value = instance.Restore(StreamIn)
```

C#

```
System.object Restore(  
    System.object StreamIn  
)
```

C++/CLI

```
System.Object^ Restore(  
& System.Object^ StreamIn  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

StreamIn

IStream interface for storage inside the SOLIDWORKS document

Visual Basic for Applications (VBA) Syntax

See [Modeler::Restore](#).

Remarks

You can use a temporary body object for display purposes. You can generate temporary bodies using such methods as [IModeler::CreateBodyFromFaces2](#) or [IModeler::ICreateBodyFromFaces3](#). Temporary bodies are not saved with the document unless you explicitly save them using [IBody2::Save](#) or [IBody2::ISave](#).

You can obtain the StreamIn object by calling [IModelDoc2::IGet3rdPartyStorage](#) and by specifying the stream name that was used during the body save operation.

See Also

SetInitKnitGapWidth Method (IModeler)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : SetInitKnitGapWidth Method (IModeler)

Sets the initial gap bound width for sewing a body. Call this method before calling any calls that attempt to knit a body; for example, [IBody2::CreateBodyFromSurfaces](#).

.NET Syntax

Visual Basic (Declaration)

```
Function SetInitKnitGapWidth( _  
    ByVal InitGapWidth As System.Double _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModeler  
Dim InitGapWidth As System.Double  
Dim value As System.Boolean  
  
value = instance.SetInitKnitGapWidth(InitGapWidth)
```

C#

```
System.bool SetInitKnitGapWidth(  
    System.double InitGapWidth  
)
```

C++/CLI

```
System.bool SetInitKnitGapWidth(  
& System.double InitGapWidth  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

InitGapWidth

Desired knitting gap width for body knitting; valid range is from 1e-8 to 0.1 meters

Return Value

True if successfully set, false if not

Visual Basic for Applications (VBA) Syntax

See [Modeler::SetInitKnitGapWidth](#).

Remarks

This initial gap width bound is adjusted during the knitting operation in an attempt to successfully knit a body. Upon completion of the knitting operation, the value is reset to the default value.

See Also

[IModeler Interface](#)
[IModeler Members](#)
[IModeler::GetInitKnitGapWidth Method](#)

SetTolerances Method (IModeler)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : SetTolerances Method (IModeler)

Obsolete. Superseded by [IModeler::GetToleranceValue](#) and [IModeler::SetToleranceValue](#).

.NET Syntax

Visual Basic (Declaration)

```
Function SetTolerances( _  
    ByRef ToleranceIDArray As System.Integer, _  
    ByRef ToleranceValueArray As System.Double, _  
    ByVal NumTol As System.Integer _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModeler  
Dim ToleranceIDArray As System.Integer  
Dim ToleranceValueArray As System.Double  
Dim NumTol As System.Integer  
Dim value As System.Boolean  
  
value = instance.SetTolerances(ToleranceIDArray, ToleranceValueArray, NumTol)
```

C#

```
System.bool SetTolerances(  
    ref System.int ToleranceIDArray,  
    ref System.double ToleranceValueArray,  
    System.int NumTol  
)
```

C++/CLI

```
System.bool SetTolerances(  
& System.int% ToleranceIDArray,  
& System.double% ToleranceValueArray,  
& System.int NumTol  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ToleranceIDArray

Type of tolerance you want to set as defined in [swTolerances](#) e

ToleranceValueArray

New tolerance value in meters for the specified tolerance type

NumTol

Original value of the specified tolerance type

Visual Basic for Applications (VBA) Syntax

See [Modeler::SetTolerances](#).

See Also

[IModeler Interface](#)

[IModeler Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

SetToleranceValue Method (IModeler)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : SetToleranceValue Method (IModeler)

Sets tolerances governing geometry output.

.NET Syntax

Visual Basic (Declaration)

```
Function SetToleranceValue( _
    ByVal ToleranceID As System.Integer, _
    ByVal NewToleranceValue As System.Double _
) As System.Double
```

Visual Basic (Usage)



```
Dim instance As IModeler
Dim ToleranceID As System.Integer
Dim NewToleranceValue As System.Double
Dim value As System.Double

value = instance.SetToleranceValue(ToleranceID, NewToleranceValue)
```

C#

```
System.double SetToleranceValue(
    System.int ToleranceID,
    System.double NewToleranceValue
)
```

C++/CLI

```
System.double SetToleranceValue(
    & System.int ToleranceID,
    & System.double NewToleranceValue
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ToleranceID

Type of tolerance you want to set as defined in [swTolerances_e](#)

NewToleranceValue

New tolerance value in meters for the specified tolerance type

Return Value

Original value of the specified tolerance type

Visual Basic for Applications (VBA) Syntax

See [Modeler::SetToleranceValue](#).

Remarks

The default tolerance setting for 3D geometry output is 0.01 meters, while 2D geometry uses a 1e-4 dimensionless tolerance value. If your 3D trim curves are coming out too far from the surfaces they are trimming or are not close to the 2D trim curves, then you might want to tighten the tolerance. Set the tolerances before any extraction of geometry.

The tolerance values can be reset by other AddIns such as IGES. Make sure that they are set to your values when you need them. The tolerance settings only affect other AddIn applications. However, you should be careful to reset the tolerances so that other AddIn applications are not affected by your choice of tolerance.

The swUVCurveOutputTol tolerance is used during the extraction of UV (SP) trim curves. The [IFace2::GetTrimCurves2](#) method is affected by this setting. This tolerance value represents a fraction of the characteristic small dimension of a face (the tolerance is dimensionless). For example, if you set the swUVCurveOutputTol tolerance value to 3e-6 and the smallest dimension of the face is 0.4 meters, then the largest deviation of the trim curve will be:

(0.4 x 3e-6) meters

For reference, the SOLIDWORKS IGES processor lets you control the swUVCurveOutputTol tolerance setting in the user preferences area. The normal setting under IGES uses a tolerance of 1e-4, while the high setting changes this tolerance to 3e-6. The IGES setting does not affect your application. Higher tolerance settings can increase processing time and the size of any output results.

See Also

[IModeler Interface](#)
[IModeler Members](#)
[IModeler::GetToleranceValue Method](#)
[IModeler::UnsetTolerances Method](#)

Availability

SOLIDWORKS 2000 SP3, Revision Number 8.3

SOLIDWORKS API Help

[Send comments on this topic.](#)

SplitFaceOnParam Method (IModeler)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : SplitFaceOnParam Method (IModeler)

Splits and retrieves the faces on the U or V parameter

Visual Basic Syntax

Visual Basic (Declaration)

```
Function SplitFaceOnParam( _  
    ByVal Facedisp As System.Object, _  
    ByVal UVFlag As System.Integer, _  
    ByVal Parameter As System.Double, _  
    ByRef Status As System.Boolean _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IModeler  
Dim Facedisp As System.Object  
Dim UVFlag As System.Integer  
Dim Parameter As System.Double  
Dim Status As System.Boolean  
Dim value As System.Object  
  
value = instance.SplitFaceOnParam(Facedisp, UVFlag, Parameter, Status)
```

C#

```
System.object SplitFaceOnParam(  
    System.object Facedisp,  
    System.int UVFlag,  
    System.double Parameter,  
    out System.bool Status  
)
```

C++/CLI

```
System.Object^ SplitFaceOnParam(  
& System.Object^ Facedisp,  
& System.int UVFlag,  
& System.double Parameter,  
& [Out] System.bool Status  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Facedisp

[Face](#) to split

UVFlag

Parametric axis; either swSplitFaceOnParamU or swSplitFaceOnParamV

Parameter

Position along the parametric axis at which the split is performed

Status

True if the operation was successful, false if

Return Value

Array of new [faces](#)

Visual Basic for Applications (VBA) Syntax

See [Modeler::SplitFaceOnParam](#).

See Also

[IModeler Interface](#)
[IModeler Members](#)
[IModeler::ISplitFaceOnParam2 Method](#)
[IModeler::ISplitFaceOnParamCount2 Method](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

ThickenSheet Method (IModeler)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : ThickenSheet Method (IModeler)

Thickens a sheet body.

.NET Syntax

Visual Basic (Declaration)

```
Function ThickenSheet( _  
    ByVal Sheet As Body2, _  
    ByVal Thickness As System.Double, _  
    ByVal Direction As System.Integer _  
) As Body2
```

Visual Basic (Usage)



```
Dim instance As IModeler  
Dim Sheet As Body2  
Dim Thickness As System.Double  
Dim Direction As System.Integer  
Dim value As Body2
```

```
value = instance.ThickenSheet(Sheet, Thickness, Direction)
```

C#

```
Body2 ThickenSheet(  
    Body2 Sheet,  
    System.double Thickness,  
    System.int Direction  
)
```

C++/CLI

```
Body2^ ThickenSheet(  
& Body2^ Sheet,  
& System.double Thickness,  
& System.int Direction  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Sheet

Sheet body that defines the profile for the temporary thickened [body](#)

Thickness

Thickness of the temporary thickened body

Direction

Direction in which to thicken the sheet body as defined in [swThickenDirection_e](#)

Return Value

Temporary thickened [body](#).

Visual Basic for Applications (VBA) Syntax

See [Modeler::ThickenSheet](#).

Example

[Thicken Sheet Body \(VBA\)](#)

Remarks

If you set Direction to swThickenDirection_Both, then the value set for Thickness is used in both directions.

See Also

[IModeler Interface](#)
[IModeler Members](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

UnsetTolerances Method (IModeler)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : UnsetTolerances Method (IModeler)

Sets the tolerances back to system settings.

.NET Syntax

Visual Basic (Declaration)

```
Function UnsetTolerances( _  
    ByRef ToleranceIDArray As System.Integer, _  
    ByVal NumTol As System.Integer _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModeler  
Dim ToleranceIDArray As System.Integer  
Dim NumTol As System.Integer  
Dim value As System.Boolean  
  
value = instance.UnsetTolerances(ToleranceIDArray, NumTol)
```

C#

```
System.bool UnsetTolerances(  
    ref System.int ToleranceIDArray,  
    System.int NumTol  
)
```

C++/CLI

```
System.bool UnsetTolerances(  
& System.int% ToleranceIDArray,  
& System.int NumTol  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ToleranceIDArray
Array specifying the tolerances to reset as defined in [swTolerances_e](#)

NumTol
Number of tolerance types you are resetting; this value should correspond to the number of elements in the ToleranceIDArray array

Return Value

True if the tolerances is reset successfully, false if n

Visual Basic for Applications (VBA) Syntax

See [Modeler::UnsetTolerances](#).

See Also

[IModeler Interface](#)
[IModeler Members](#)
[IModeler::GetToleranceValue Method](#)
[IModeler::SetToleranceValue Method](#)